

31 10 00 - Site Clearing

All unsuitable materials, as defined in a Soils Engineering Report, shall be removed from the site and disposed of in a suitable location.

Erosion control measures and BMPs shall be implemented in accordance to the SWPPP and dust control shall be employed at all times.

31 22 00 - Grading

Ponding and areas of low flow gradients should be avoided unless part of a designed storm water mitigation measure such as bio-retention swales, or rain gardens. Positive surface drainage or drainage devices such as swales or drain inlets shall be provided wherever required to convey water away from building foundations.

Storm drainage shall be directed away from all buildings. If in the rare case this cannot be accomplished, redundant drainage systems are required. There shall be no potential for ponding water that could flood into a building. Safe overland flow paths must be provided in case storm drains clog or their capacity is exceeded.

Site retaining or planter walls shall have waterproofing and adequate drainage to prevent additional loading. Provide a water proofing system at the backside, and foundation drains at the base. Drains may be tied into the storm drain or an appropriate location. Retaining walls more than 30" in height from top to finished grade are required to have a 42" high guardrail or other approved protective measure. Provide flat areas for all utility slabs, vaults or boxes.

Finished grade elevations shown on the grading plan shall be to the finished surface elevation at 0.1' increments for soils surfaces and at 0.01' increments for hard finish surfaces. Contractor shall subtract structural thickness of pavements of top soil to achieve rough grading elevations. Proposed grading contours shall be indicated at 1-foot contour intervals.

31 23 33 - Trenching and Backfilling

Standard: Coordinate backfilling with installation of underground duct markers for utilities. Backfill with sand to 12 inches above utility duct. Install duct markers 6 inches below finished surface.

Schedule: Storm and Sanitary Piping: Cover pipe and bedding with inches

Utilities	Fill Type	Bedding and Cover Thickness, minimum	Compaction, minimum	Underground Duct Markers
Water Utilities Including Irrigation	Sand	1" below 3" above		
Sanitary Sewerage Utilities				
Storm Drainage Utilities				

Natural-Gas Distribution Utilities				
Hydronic Utilities			95%	
Electrical Utilities				
Communication			95%	